Figure 1.1 The differences between the LLM interactions from direct action compared to using proxy agents, agents, and autonomous agents

Figure 1.1 demonstrates the use cases for a single flow of actions on an LLM using a single agent. For more complex problems, we often break agents into profiles or personas. Each agent profile is given a specific task and executes that task with specialized tools and knowledge.

Multi-agent systems are agent profiles that work together in various configurations to solve a problem. Figure 1.2 demonstrates an example of a multi-agent system using three agents: a controller or proxy and two profile agents as workers controlled by the proxy. The coder profile on the left writes the code the user requests; on the right is a tester profile designed to write unit tests. These agents work and communicate together until they are happy with the code and then pass it on to the user.

Figure 1.2 shows one of the possibly infinite agent configurations. (In chapter 4, we'll explore Microsoft's open source platform, AutoGen, which supports multiple configurations for employing multi-agent systems.)